

KIRIKKALE, Turkey

WMO: 171350

Lat: 39.8433N

Lon: 33.5181E

Elev: 748

StdP: 92.66

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-7.4	-13.3	1.3	-8.0	-11.1	1.6	-5.5	6.4	1.8	5.5	1.9	1.5	80	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	35.8	18.6	34.1	18.2	32.4	18.0	21.8	31.1	20.5	29.6	19.4	28.6	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	14.8	27.4	17.2	13.5	26.0	16.0	12.4	24.5	67.5	31.3	62.7	29.6	58.7	28.6	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.4	4.7	-11.6	38.8	4.0	1.7	-14.5	40.0	-16.8	41.0	-19.1	42.0	-22.0	43.2		
(5)			-12.2	21.5	3.9	2.7	-15.0	23.4	-17.2	25.0	-19.4	26.5	-22.2	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	0.9	3.7	8.3	13.3	16.8	21.3	25.2	25.1	20.6	14.1	7.8	2.9		
	DBStd	9.03	4.64	4.78	3.99	3.37	3.08	2.87	2.55	2.77	3.42	3.69	3.57	3.81		
	HDD10.0	860	283	177	79	8	0	0	0	0	0	8	83	222		
	HDD18.3	2440	541	408	313	155	67	7	0	1	17	137	316	479		
	CDD10.0	2094	0	2	24	107	212	338	472	468	317	137	17	1		
	CDD18.3	633	0	0	1	3	20	94	213	210	85	7	0	0		
	CDH23.3	7216	0	0	5	73	248	1002	2484	2396	912	96	0	0		
	CDH26.7	3164	0	0	0	9	51	369	1211	1158	349	18	0	0		
(14) Wind	WSAvg	1.8	1.4	1.7	1.8	1.8	1.7	2.0	2.4	2.5	1.7	1.4	1.3	1.3		
(15) Precipitation	PrecAvg	382	44	32	32	45	54	35	12	8	15	28	30	48		
	PrecMax	599	88	90	63	101	100	112	53	27	62	83	115	103		
	PrecMin	245	2	2	6	4	7	2	0	0	0	0	0	9		
	PrecStd	87	26	22	17	25	24	26	14	7	17	23	26	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	17.9	22.9	28.2	30.4	34.5	38.6	37.8	35.1	29.1	20.2	13.1		
		MCWB	8.2	9.6	11.8	13.3	15.4	17.5	18.8	17.9	15.8	17.1	11.6	8.9		
	2%	DB	10.2	14.5	19.7	25.4	28.2	32.5	36.3	35.7	32.5	26.0	18.0	11.0		
		MCWB	6.8	8.2	10.4	12.8	14.8	17.5	19.3	18.4	16.7	14.4	10.8	7.9		
	5%	DB	8.4	12.5	17.6	23.1	26.3	30.6	34.3	34.0	30.5	23.7	16.1	9.7		
		MCWB	5.9	7.4	9.5	12.5	14.8	16.9	18.9	18.5	16.6	13.8	10.3	7.3		
	10%	DB	6.7	10.5	15.4	21.0	24.3	28.7	32.3	32.3	28.5	21.2	14.2	8.3		
		MCWB	5.0	6.6	8.7	11.7	14.5	16.5	18.2	18.3	16.2	13.0	9.2	6.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	10.2	13.2	15.6	18.0	21.1	24.2	23.2	20.2	19.0	13.1	10.0		
		MCDB	11.7	15.8	20.5	23.1	25.2	30.4	34.4	31.5	29.2	27.5	18.4	11.5		
	2%	WB	7.2	8.8	11.3	14.2	16.6	19.4	22.2	21.7	19.1	15.7	11.7	8.6		
		MCDB	9.6	13.3	17.8	22.0	24.2	27.7	31.8	30.9	28.6	22.9	16.3	10.5		
	5%	WB	6.1	7.9	10.1	13.2	15.7	18.2	20.8	20.6	17.9	14.5	10.6	7.6		
		MCDB	8.2	11.6	16.1	21.1	23.5	27.2	30.0	29.4	26.7	20.8	14.8	9.3		
	10%	WB	5.1	6.8	9.1	12.2	15.0	17.3	19.4	19.4	16.8	13.7	9.6	6.5		
		MCDB	6.7	10.2	14.7	19.6	22.6	26.3	28.7	28.8	25.3	19.7	13.5	8.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	8.6	10.8	12.3	12.2	12.9	13.6	13.3	13.6	11.9	11.0	6.2		
		MCDBR	8.5	11.9	14.7	16.1	15.2	15.6	15.8	15.3	16.6	15.6	13.4	8.0		
	5% WB	MCWBR	5.6	6.7	7.6	6.9	5.7	5.1	5.3	4.7	6.4	7.4	7.5	5.3		
		MCWBR	7.8	10.4	12.8	14.2	13.5	13.6	14.3	13.3	14.5	13.0	11.7	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.343	0.387	0.428	0.404	0.381	0.380	0.378	0.356	0.359	0.325	0.315		
	taud		2.356	2.293	2.189	2.104	2.244	2.334	2.336	2.318	2.379	2.358	2.422	2.411		
	Ebn at Noon		845	866	860	847	875	894	891	884	885	838	828	816		
	Edh at Noon		87	107	133	154	137	125	124	123	108	99	81	77		
(44) All-Sky Solar Radiation	RadAvg		1.88	2.91	4.07	5.21	6.18	7.08	7.50	6.70	5.28	3.61	2.48	1.66		
	RadStd		0.23	0.43	0.35	0.48	0.45	0.54	0.30	0.25	0.35	0.26	0.23	0.21		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page